Joint Distributions

Take X and Y two random variables, such that:

$$P(x, y) = P(X = x, Y = y)$$

The PDF is defined as $f(x, y)$ and the probability that we observe an event inside a given 2D area A is:

$$P((x, y) \in A) = \int_A f(x, y) dx dy$$

Independence

The two variables are independent if

$$P(X = x, Y = y) = P(X = x) \times P(Y = y)$$

Which can be used for separation of variables.

One point to be considered, in variable decomposition, is that some coordinate systems are more suitable, in a given scenario, to variable separation as opposed to others.

Marginal Distribution

A marginal distribution considers a subset of variables:

> What distribution would I get for X if I ignored Y?

For $P(X, Y)$, we have marginals $P(X)$ and $P(Y)$.

For discrete values, we sum out the others:

$$P(X = x) = \sum_{y \in Y} P(X = x, Y = y)$$

For continuous values, we integrate out the others:

$$p(x) = \int_{-\infty}^{\infty} p(x, y) dy$$

Conditional Distribution

A conditional distribution fixes a subset of variables.

> What distribution would I get for X if I set $Y = y$?

For both discrete and continuous, divide the joint distribution by the marginal.

$$P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$$

Where the denominator is the marginal distribution we saw earlier (sum or integral).

This can be defined as the **Product Rule**:

$$P(X = x, Y = y) = P(X = x|Y = y) \times P(Y = y)$$

Which is the basis for the **Bayes' Rule**:

$$P(X = x|Y = y) = \frac{P(Y = y|X = x) \times P(X = x)}{P(Y = y)}$$

References

- Basics of joint probability
- Joint Probability Distributions
- Joint Probability Distributions: Marginal and Conditional Densities